

# Project Design Phase-II

## Technology Stack (Architecture & Stack)

|               |                                                          |
|---------------|----------------------------------------------------------|
| Date          | 29 September 2026                                        |
| Team ID       | SWTID-2026-2603                                          |
| Project Name  | AnswerBot – AI-Powered FAQ & Customer Support Automation |
| Maximum Marks | 4 Marks                                                  |

### Technical Architecture

AnswerBot uses a REST API architecture. The client sends requests to a Node.js/Express server. Authentication and application services process the request, MongoDB stores application data, and Google Gemini is used when AI FAQ generation is requested.

|                                            |                                                                |                                      |
|--------------------------------------------|----------------------------------------------------------------|--------------------------------------|
| Client / User Interface                    | REST API                                                       | Data & AI Services                   |
| Web / API client Postman or application UI | Node.js + Express.js JWT authentication Controllers / Services | MongoDB + Mongoose Google Gemini API |

External interface: Google Gemini API. Data storage: MongoDB. Security boundary: JWT-protected API routes with bcrypt password hashing and schema validation.

### Table-1: Components & Technologies

| S.No | Component                       | Description                                                         | Technology                                |
|------|---------------------------------|---------------------------------------------------------------------|-------------------------------------------|
| 1    | User Interface                  | Application interface/API client used to interact with AnswerBot.   | REST API / Web UI                         |
| 2    | Application Logic-1             | Authentication, routing and request handling.                       | Node.js + Express.js                      |
| 3    | Application Logic-2             | FAQ CRUD, search, categories and validation.                        | Express.js controllers/services           |
| 4    | Application Logic-3             | AI-assisted FAQ generation.                                         | Google Gemini                             |
| 5    | Database                        | User, FAQ and category data.                                        | MongoDB + Mongoose                        |
| 6    | Cloud Database                  | Optional managed database deployment.                               | MongoDB Atlas / managed MongoDB           |
| 7    | File Storage                    | No dedicated file-storage component required for core FAQ workflow. | Local filesystem / optional cloud storage |
| 8    | External API-1                  | AI FAQ generation.                                                  | Google Gemini API                         |
| 9    | External API-2                  | No additional external API required for the core solution.          | —                                         |
| 10   | Machine Learning Model          | Generative AI used to draft FAQ content from supplied prompts.      | Google Gemini model                       |
| 11   | Infrastructure (Server / Cloud) | Application deployment environment.                                 | Local Server / Cloud deployment           |

### Table-2: Application Characteristics

| S.No | Characteristics          | Description                                                                                                                     | Technology                                  |
|------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 1    | Open-Source Frameworks   | Backend framework and database ODM used to build the REST service.                                                              | Node.js, Express.js, Mongoose               |
| 2    | Security Implementations | Authentication, password protection, route protection, validation and role-based access.                                        | JWT, bcrypt, schema validation, RBAC        |
| 3    | Scalable Architecture    | Modular REST API separates routing, authentication, services and persistence.                                                   | Node.js + Express.js + MongoDB              |
| 4    | Availability             | Can be deployed locally or to a suitable cloud environment; centralized error handling supports graceful failures.              | Cloud/Local deployment + API error handling |
| 5    | Performance              | Indexed/searchable FAQ data and API-based processing reduce unnecessary work; AI calls occur only when generation is requested. | MongoDB + Express.js + Gemini API           |

### Team Members

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